

Platform

Solutions

Resources

Open Source

Enterprise

Pricing

Search or jump to...

7

Sign in

Sign up

ImL1s/antigravity_for_loop

Public

Notifications

Fork 5

Star 24

<> Code

Issues

Pull requests 1

Actions

Projects

Security and quality

Insights

main1 Branch3 Tags

Go to file

<> Code

yamiweb3 chore: bump version to 0.2.1 and release

a31688 · 5 months ago

14 Commits

.agent/workflows

feat: Add CDP support for Antigravity iframe + Lexical editor

5 months ago

.github/workflows

fix: Switch E2E tests from wdio-vscode-service to @vscode/te...

5 months ago

.playwright-mcp

fix: Switch E2E tests from wdio-vscode-service to @vscode/te...

5 months ago

assets

feat: Add comprehensive test suite (40 tests) and Status Bar UX

5 months ago

commands

feat: Add CDP support for Antigravity iframe + Lexical editor

5 months ago

docs

feat: Add CDP support for Antigravity iframe + Lexical editor

5 months ago

lib

feat: Implement Ralph Loop architecture (inspired by Claude ...

5 months ago

reference_repos

fix: resolve E2E test timeout in extension.e2e.test.js

5 months ago

spec

feat: Add CDP support for Antigravity iframe + Lexical editor

5 months ago

test_project

fix: resolve E2E test timeout in extension.e2e.test.js

5 months ago

test-results

fix: Switch E2E tests from wdio-vscode-service to @vscode/te...

5 months ago

test

fix: resolve E2E test timeout in extension.e2e.test.js

5 months ago

.gitignore

feat: Add comprehensive test suite (40 tests) and Status Bar UX

5 months ago

.shellspec

Initial commit: Antigravity For Loop plugin with tests and CI/...

5 months ago

.vscode-test.mjs

fix: Switch E2E tests from wdio-vscode-service to @vscode/te...

5 months ago

.vscodeignore

feat: Add CDP support for Antigravity iframe + Lexical editor

5 months ago

AGENTS.md

feat: Add CDP support for Antigravity iframe + Lexical editor

5 months ago

LICENSE

Add LICENSE and fix publisher name for Open VSX

5 months ago

README.md

docs: Update README with accurate test counts and languag...

5 months ago

extension.js

docs: Update README with accurate test counts and languag...

5 months ago

firebase-debug.log

fix: resolve E2E test timeout in extension.e2e.test.js

5 months ago

latest_run.log

fix: resolve E2E test timeout in extension.e2e.test.js

5 months ago

package-lock.json

chore: bump version to 0.2.1 and release

5 months ago

package.json

chore: bump version to 0.2.1 and release

5 months ago

test-iframe.js

fix: resolve E2E test timeout in extension.e2e.test.js

5 months ago

wdio.conf.js

fix: Switch E2E tests from wdio-vscode-service to @vscode/te...

5 months ago

README

MIT license

Antigravity For Loop

Ralph Loop for Antigravity - Autonomous AI development loop that keeps the AI agent working until tests pass or max iterations reached.

Inspired by Claude Code's Ralph Wiggum plugin, redesigned for Google Antigravity IDE.

How It Works

RALPH LOOP FOR ANTIGRAVITY

Inject Prompt

Submit + Auto-Accept

Wait for AI Complete

Check Completion

- Test command exit code == 0?
- AI output contains "DONE"?

iteration < max_iterations?

NO

YES

DONE

vs Claude Code Ralph Wiggum

Aspect	Claude Code	Antigravity For Loop
Interface	CLI	GUI (VSCode-like)
Loop Mechanism	Stop Hook intercepts exit	CDP + setInterval
Prompt Injection	Hook re-injects prompt	CDP → Lexical Editor
Completion Detection	Hook exit code 2	Test exit code 0 / "DONE"
Auto-Accept	Not needed (CLI)	CDP clicks Accept buttons

Features

Ralph Loop = Inject Prompt → Wait for AI → Run Tests → Repeat

About

自動迭代修復循環插件: 讓 AI 代理持續執行開發任務直到通過所有測試

Readme

MIT license

Activity

24 stars

0 watching

5 forks

Report repository

Releases

3 tags

Packages

No packages published

Contributors 2

yamiweb3

claude Claude

Languages

JavaScript 87.5%

Shell 12.5%

- **Ralph Loop** - inject prompt -> wait for AI -> run tests -> repeat
- **Auto-Accept** - Automatically click Accept/Run buttons via CDP
- **Smart Detection** - Auto-detect test commands for 25+ languages
- **Real-time Progress** - Status bar shows iteration progress
- **Safety Limits** - `maxIterations` prevents infinite loops

Installation

```
# Package the extension
npm run package

# Install in Antigravity IDE
code --install-extension antigravity-for-loop-*.vsix
```

CDP Setup (Required)

Antigravity must have CDP (Chrome DevTools Protocol) enabled for the extension to work.

Automatic Setup (Recommended)

1. After installing the extension, a prompt will appear
2. Click "Enable CDP"
3. Follow instructions to restart Antigravity

Manual Setup

macOS:

```
open -a "Antigravity.app" --args --remote-debugging-port=9000
```

Windows:

```
Add to shortcut target: --remote-debugging-port=9000
```

Linux:

```
antigravity --remote-debugging-port=9000
```

Usage

Quick Start

1. Click "For Loop" button in the status bar
2. Select "Start Ralph Loop..."
3. Enter task description (e.g., "Fix all TypeScript errors")
4. Select completion condition (Tests Pass / Build Succeeds / AI Self-Judgment)
5. Select maximum iterations
6. Go!

Keyboard Shortcuts

Shortcut	Function
<code>Cmd+Alt+Shift+L</code>	Open menu
<code>Cmd+Alt+Shift+A</code>	Toggle Auto-Accept
<code>Cmd+Alt+Shift+C</code>	Copy Continuation Prompt

Completion Conditions

Option	Description
Tests Pass	Auto-detect test command, stop on exit code 0
Build Succeeds	Stop when build completes successfully
AI Self-Judgment	Stop when AI outputs "DONE"
Custom Command	Enter a custom validation command

Supported Languages (Auto-Detection)

The extension automatically detects test commands for 25+ languages and build systems:

Category	Languages/Frameworks
JavaScript	npm, Deno, Bun
Python	pytest, tox, mypy
Systems	Rust (cargo), Go, Zig, Nim, V
JVM	Java/Kotlin (Maven, Gradle), Scala (sbt), Clojure (lein)
Functional	Haskell (stack, cabal), Elixir (mix), Erlang (rebar3), OCaml (dune)
Mobile	Swift, Dart/Flutter
Web	Ruby (rspec), PHP (phpunit), .NET (dotnet)
C/C++	CMake, Meson, Conan
Build Systems	Make, Just, Task, Bazel, Pants, Earthly

Testing

```
# Run all unit tests (CI/CD compatible)
npm test

# Run specific test suites
npm run test:unit:ralph # RalphLoop tests
npm run test:unit:cdp   # CDPManager tests
npm run test:unit:state # State parser tests
npm run test:unit:detect # Test command detection

# Run CDP E2E tests (local only, requires Antigravity IDE)
npm run test:e2e:cdp
```

Test Coverage

Test Suite	Tests	CI/CD	Description
ralphLoop.test.js	37	✓	Loop logic, prompt building, callbacks
cdpManager.test.js	29	✓	CDP connection, helper script validation
stateParser.test.js	14	✓	State parsing, status bar text
detectTestCommand.test.js	11	✓	Test command detection for core languages
cdp-ralph-loop.e2e.js	16	✗	Real CDP E2E (requires Antigravity IDE)

Total: 91 unit tests + 16 E2E tests = 107 tests

Running CDP E2E Tests

These tests require a real Antigravity IDE instance:

```
# 1. Start Antigravity with CDP enabled
open -a "Antigravity.app" --args --remote-debugging-port=9000

# 2. Open the Agent Panel in Antigravity

# 3. Run E2E tests
npm run test:e2e:cdp
```

Directory Structure

```
antigravity_for_loop/
├── extension.js           # VSCode extension entry point
├── package.json          # Extension manifest & scripts
├── lib/
│   ├── cdp-manager.js    # CDP connection & DOM injection
│   ├── ralph-loop.js     # Ralph Loop core logic
│   └── relauncher.js     # CDP restart helper
├── test/
│   ├── unit/             # Unit tests (CI/CD compatible)
│   │   ├── ralphLoop.test.js
│   │   ├── cdpManager.test.js
│   │   ├── stateParser.test.js
│   │   └── detectTestCommand.test.js
│   ├── e2e/              # E2E tests (local only)
│   │   └── cdp-ralph-loop.e2e.js
│   ├── integration/      # VSCode integration tests
│   │   └── extension.test.js
├── assets/
│   └── icon.png
└── README.md
```

Technical Details

CDP Architecture

The extension uses Chrome DevTools Protocol to interact with Antigravity's webview:

1. Connect to CDP on port 9000 (scans 9000-9003)
2. Find the `antigravity.agentPanel` iframe
3. Locate the Lexical editor (`[data-lexical-editor="true"]`)
4. Inject text using `execCommand("insertText")`
5. Click Submit and Accept buttons via DOM queries

Helper Script Functions

```
window.__antigravityForLoop = {
  findChatInput() // Find Lexical editor in iframe
  findSubmitButton() // Find Submit button
  injectPrompt(text) // Inject text into editor
  submitPrompt() // Click Submit or press Enter
  isAcceptButton(el) // Check if element is Accept button
  clickAcceptButtons() // Click all Accept buttons
  getAIStatus() // Get AI response status
}
```

Ralph Loop Lifecycle

```
1. Start Loop
  ↳ Connect to CDP
  ↳ Inject helper script
  ↳ Start auto-accept interval

2. Each Iteration
  ↳ Build prompt with context
  ↳ Inject prompt into Lexical editor
  ↳ Submit prompt
  ↳ Wait for AI completion
  ↳ Run test command
  ↳ Check success/failure

3. Completion
  ↳ Tests pass (exit 0) → SUCCESS
  ↳ AI says "DONE" → SUCCESS
  ↳ Max iterations reached → FAILURE
  ↳ User cancels → CANCELLED
```

Security Notes

- Recommend `git commit` before using
- Use `maxIterations` to limit iterations (default: 10)
- Manually confirm high-risk operations (e.g., file deletion)
- CDP requires port 9000, ensure no conflicts

License

MIT License

Made for Google Antigravity IDE

Inspired by [Claude Code](#) [Naïm Waggum](#)